

BIBEK PATHAK

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SUMMARY

Information Technology student building real-time computer-vision and full-stack systems in Python, TypeScript, and SQL. Hands-on work spans an autonomous-driving vision pipeline, a recommendation engine, and data-driven web apps, backed by Advanced SQL and Tableau. Seeking a SWE, AI/ML, or data internship.

TECHNICAL SKILLS

Languages: Python, SQL, TypeScript, JavaScript, Java

AI / ML: Computer Vision, YOLOv8, OpenCV, scikit-learn, XGBoost/LightGBM, Anthropic Claude API, recommendation systems

Data: SQL (Advanced), PostgreSQL, MongoDB, Tableau, Excel, data quality and reconciliation

Backend: FastAPI, Node.js, Express, REST APIs, JWT auth, Docker

Frontend: React, Next.js, Vite, Tailwind CSS | Tools: Git, GitHub, Vercel, Render, pytest, Linux

PROJECTS

Cruze [\[In Progress\]](#) | Python, YOLOv8, OpenCV, faster-whisper, Piper, Claude API

[GitHub](#)

- Architected a vision-first driving co-pilot as an async event-driven pub/sub pipeline (**camera** → **detection** → **depth** → **tracking** → **scene** → **rules** → **voice**), so perception modules stay fully decoupled and independently testable
- Implemented real-time threat detection with explicit math (forward-collision time-to-collision under 3s, tailgating headway, speeding, stop-sign and slow-lead alerts) running on a simulated camera feed on desktop
- Designed a hardware-abstraction layer that swaps YOLOv8 CPU, TensorRT (Jetson), and TFLite (Raspberry Pi + Hailo) backends via a one-line config change, plus an offline voice stack with Claude API dialog

VectorVance | Python, Computer Vision, Pathfinding

[GitHub](#)

- Engineered an autonomous car simulation in Python with real-time vector-based navigation, pathfinding, and obstacle-avoidance logic that steers the vehicle around dynamic hazards

WatchThis! AI [\[In Progress\]](#) | Next.js, FastAPI, PostgreSQL, Docker

[GitHub](#)

- Building a movie and TV recommendation engine that returns explainable recommendations, served by a Python FastAPI backend over PostgreSQL and containerized with Docker for reproducible deploys

Car Deal Finder [\[In Progress\]](#) | FastAPI, React, PostgreSQL, XGBoost/LightGBM

[GitHub](#)

- Building a full-stack tool that surfaces undervalued car listings by scraping marketplace data and scoring deals against an ML price-estimation model (XGBoost/LightGBM) behind a FastAPI + React interface

KaryaAI | React, Node.js, Express, MongoDB, JWT

[Live](#)

- Built and shipped a full-stack AI task manager to production with JWT auth, session persistence, and real-time MongoDB syncing, including AI-driven task prioritization and a responsive custom UI

GridNavigator | JavaScript, BFS, DFS

[Live](#)

- Built an interactive maze generator that visualizes BFS and DFS pathfinding step-by-step with animated traversal and theme switching, deployed live

EXPERIENCE

IT Student Worker

Sep 2025 – Present

SLU College of Science & Technology

Hammond, LA

- Maintained departmental records for 200+ student interactions using Excel and SQL, restructuring the data so staff could retrieve student records faster
- Resolved first-line database and software issues for students and faculty, and coordinated logistics for 3+ campus engagement events

Junior Accountant

Jan 2023 – Mar 2024

OM Casting Udhyog Pvt. Ltd.

Birtamode, Nepal

- Processed monthly payroll for 200+ employees and ran bank reconciliations, ensuring accurate, on-time financial statements and labor-law compliance
- Performed data-quality reviews on financial reports, flagging discrepancies that fed into business strategy planning

EDUCATION

Southeastern Louisiana University

Hammond, LA

B.S. Information Technology — Honors Scholarship recipient

Expected May 2028

Relevant coursework: Data Structures & Algorithms, Software Engineering, Database Systems, Discrete Structures, Calculus

CERTIFICATIONS

HackerRank SQL (Advanced) • HackerRank Software Engineer • CFI Excel: Formulas for Finance • Simplilearn Intro to Tableau